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so the effect of I_3 is a renormalization of the Σ_ϵ coefficient at this order.

For compactness in later sections we will also use the invariant $I_T \equiv -\frac{1}{4}\,\Pi_{\rm PT}\big[T^A{}_{BC}T_A{}^{BC}\big] = -6\,\eta^2\,\Sigma_\epsilon$ (which follows after C1 in Section 4), alongside the canonical traceless matrix $\mathcal{T}$ and scale τ defined in (8)–(9).

Null Test B (stated; paper-checkable).

On any admissible background with the fall-offs of A4, every projected, PT-even quadratic density with at most one derivative per building block reduces *pre-lock* to $c_1 I_1 + c_2 I_2 + c_3 I_3$ up to a total derivative, and *post-lock* to $c_1 I_1 + \tilde{c}_2 \Sigma_\epsilon$ up to a total derivative. A constructive IBP reduction appears in the Appendix.

Order of Operations (Projection ↔ Variation; Lock vs. Closure)

The scalar projector is applied at the level of *densities*; by Lemma 1 (Section 2) *project-then-vary* and *vary-then-project* commute on scalar densities. The trace lock (Equation (17)) is an *algebraic* constraint introduced at the level of equations of motion or via a Lagrange current; it is not a PT projection. Accordingly, the correct sequence for building the quadratic sector is:

(i) enumerate & project $\Rightarrow$ pre-lock basis $\{I_1, I_2, I_3\}$ $\quad\Longrightarrow\quad$ (ii) impose trace lock/C1 $\Rightarrow I_3 \to 3\eta\,\Sigma_\epsilon$.

This sequencing will be used verbatim in Sections 4–5.

4. Uniqueness Theorem (C1)

Under the posture of Sections 2–3 (A1–A5, scalar PT projector, "one derivative per building block"), algebraic torsion consistent with the projector is *uniquely* pure trace and aligned with the spurion gradient:

$$\boxed{T^A{}_{BC} = 2\eta\,\delta^A{}_{[B}\,\partial_{C]}\epsilon, \qquad \eta > 0,} \tag{13}$$

so that the axial and traceless irreps vanish, $S^\mu{=}0$ and $q_{\lambda\mu\nu}{=}0$. Equivalently,

$$\Pi_{\rm PT}\big[T^A{}_{BC}T_A{}^{BC}\big] = \frac{2}{3}\,T_\mu T^\mu = 6\,\eta^2\,\Sigma_\epsilon \quad\Longrightarrow\quad \boxed{I_T \equiv -\tfrac{1}{4}\,\Pi_{\rm PT}[T^A{}_{BC}T_A{}^{BC}] = -6\,\eta^2\,\Sigma_\epsilon.} \tag{14}$$

Here $\Sigma_\epsilon \equiv \Pi_{\rm PT}\big[(\partial\epsilon)^2\big]$ is the *projected*, *real* scalar, and we use the shorthands from Section 3.1: $\mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon)$, $\hat{n}_\mu \equiv \partial_\mu\epsilon/\sqrt{|\Sigma_\epsilon|}$, $\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat{n}^\mu\hat{n}_\nu - \frac{1}{4}\delta^\mu{}_\nu\Big)$, and $\tau \equiv \hat{n}^\mu T_\mu = 3\eta\ \mathrm{sgn}(\Sigma_\epsilon)\,\sqrt{|\Sigma_\epsilon|}$.

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